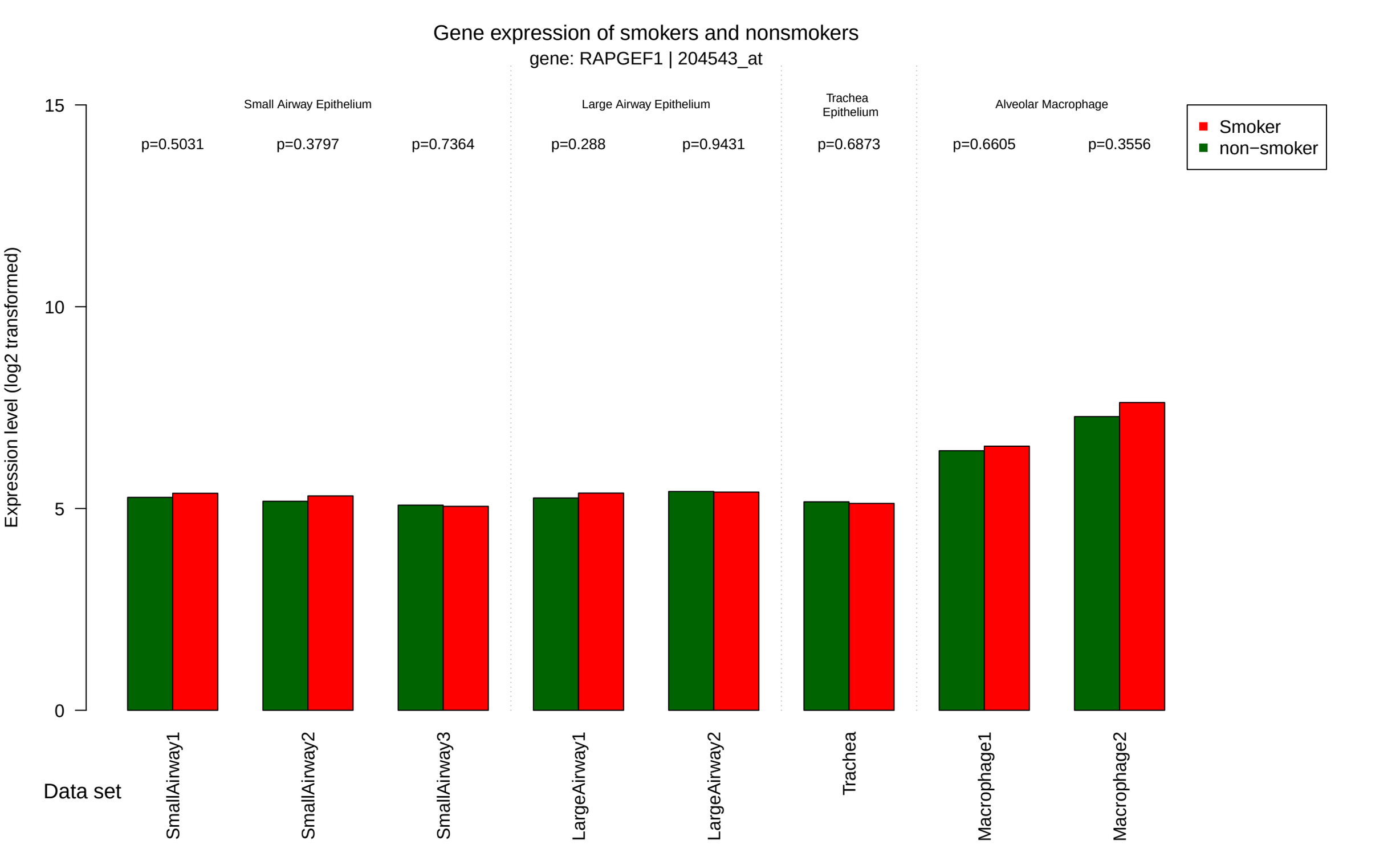
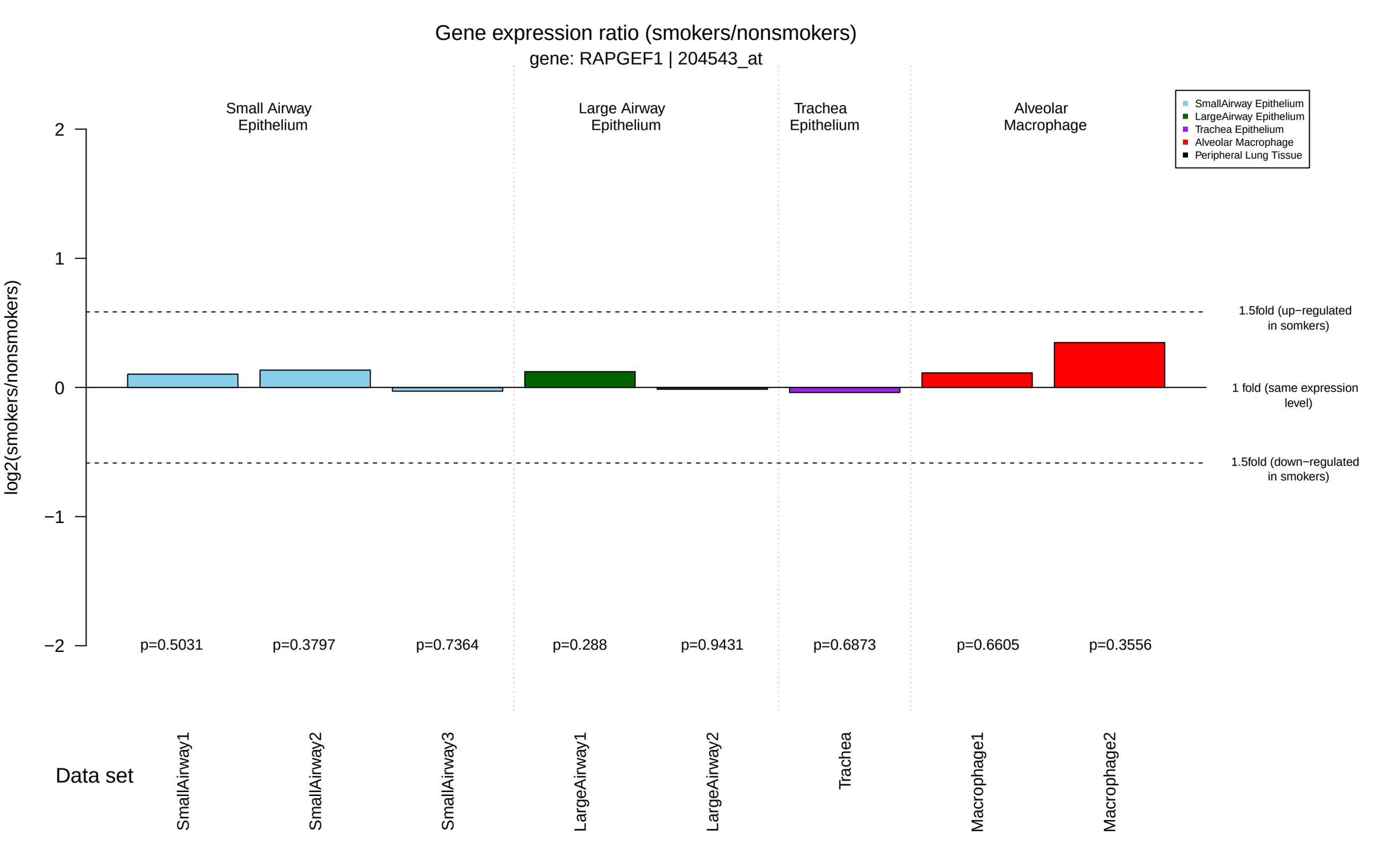
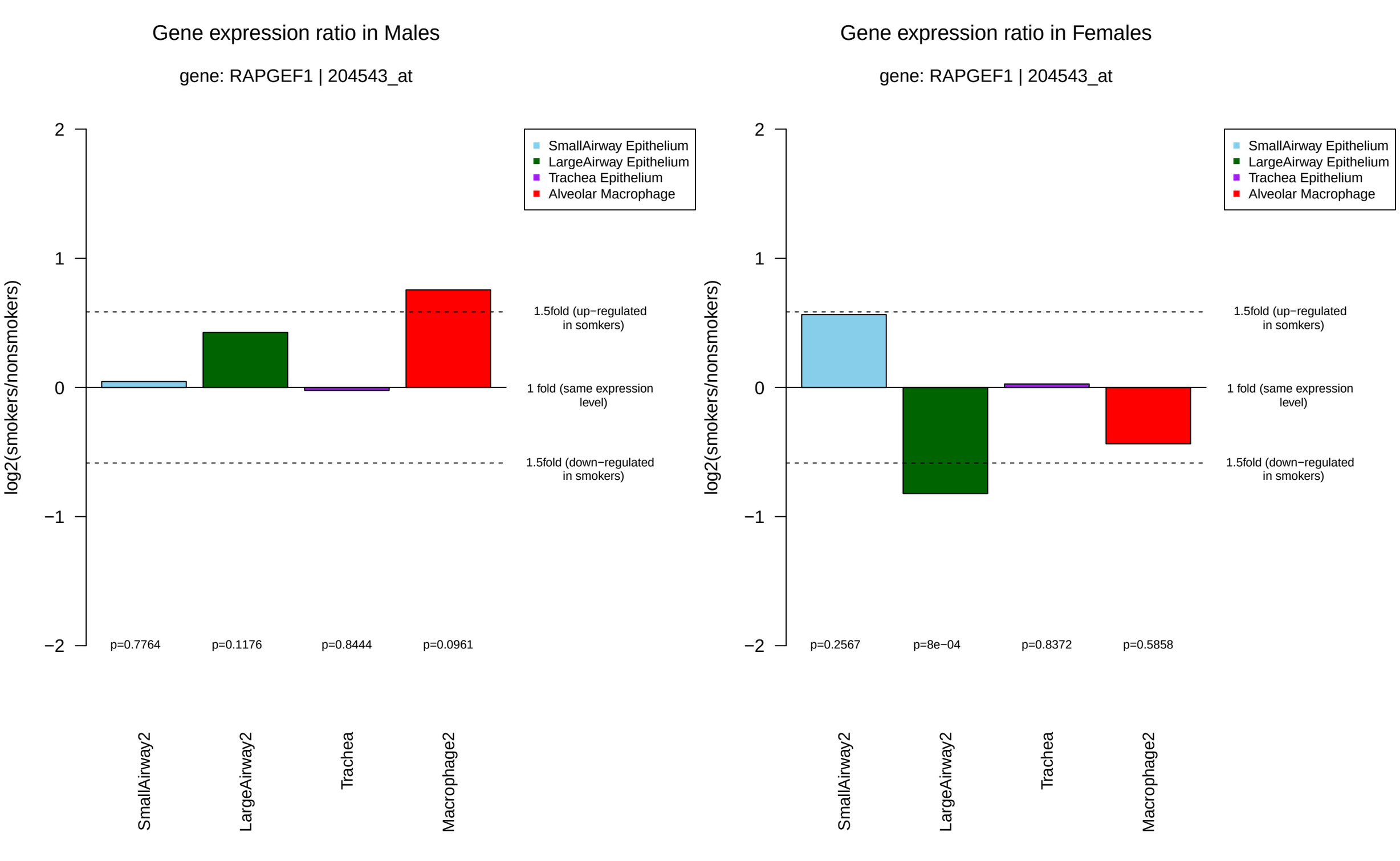
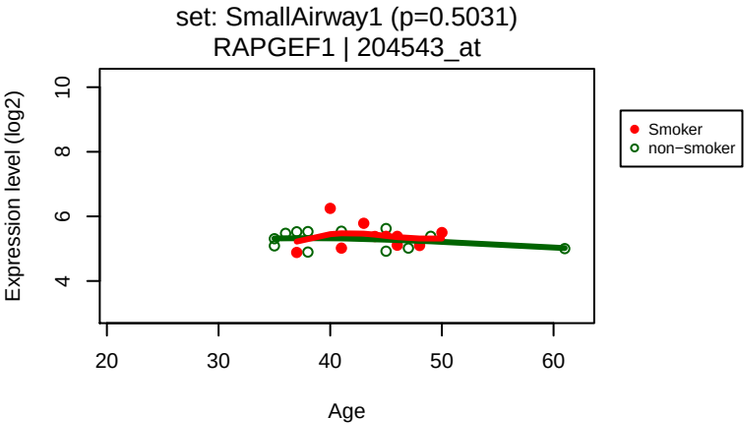
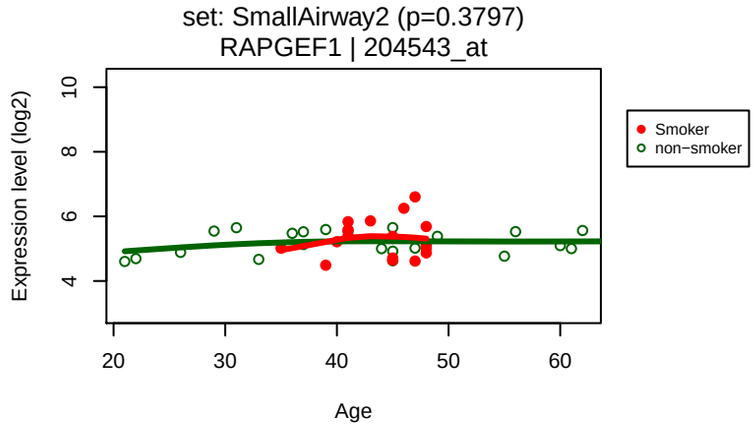
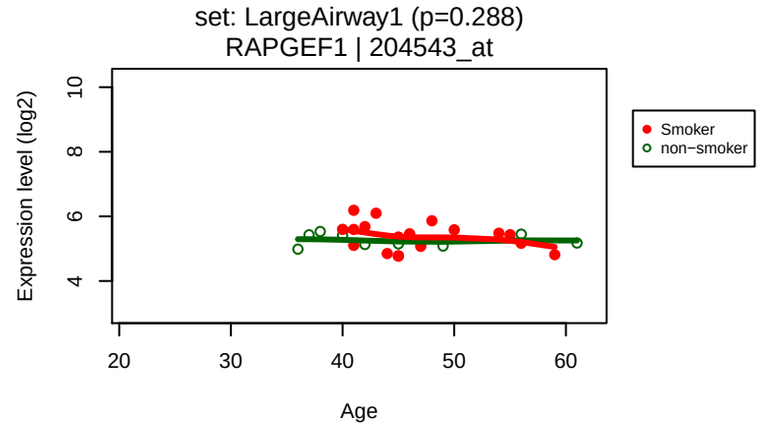

Gene expression of smokers and nonsmokers
gene: RAPGEF1 | 204543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5031
p=0.3797
p=0.7364
p=0.288
p=0.9431
p=0.6873
p=0.6605
p=0.3556
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

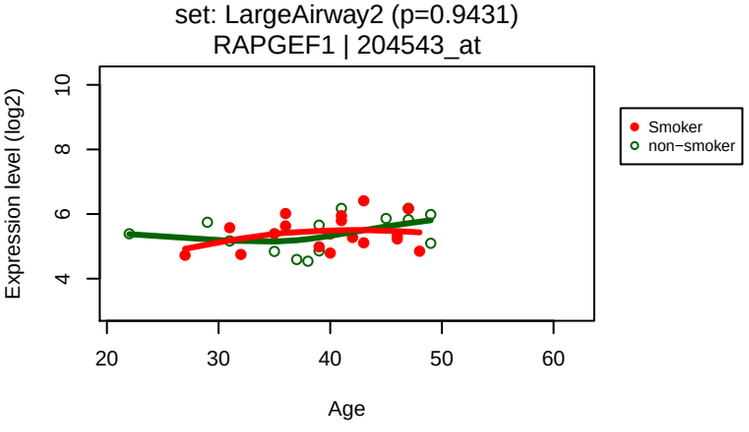

Gene expression ratio (smokers/nonsmokers)
gene: RAPGEF1 | 204543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5031
p=0.3797
p=0.7364
p=0.288
p=0.9431
p=0.6873
p=0.6605
p=0.3556
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

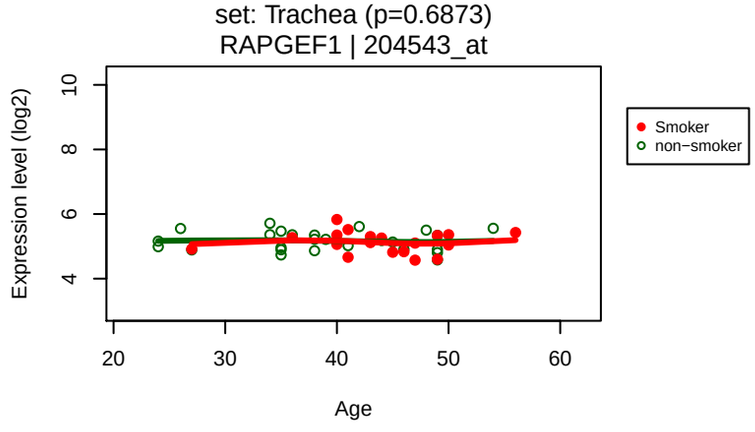

Gene expression ratio in Males
gene: RAPGEF1 | 204543_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7764
p=0.1176
p=0.8444
p=0.0961
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAPGEF1 | 204543_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2567
p=8e−04
p=0.8372
p=0.5858
SmallAirway2
LargeAirway2
Trachea
Macrophage2

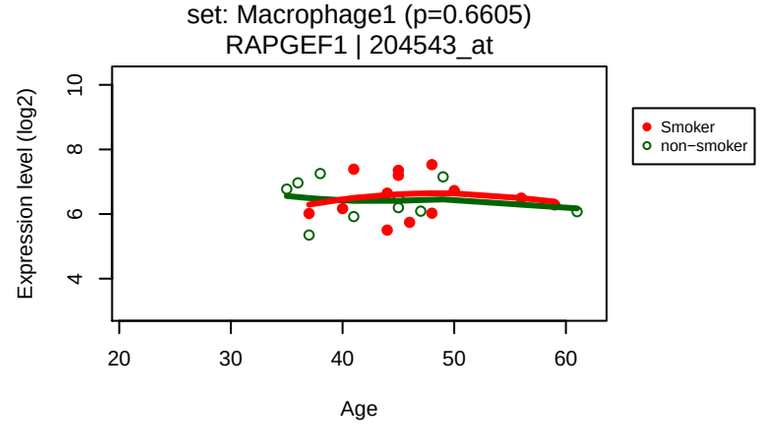

set: SmallAirway1 (p=0.5031)
RAPGEF1 | 204543_at
Expression level (log2)
Age
Smoker
non−smoker

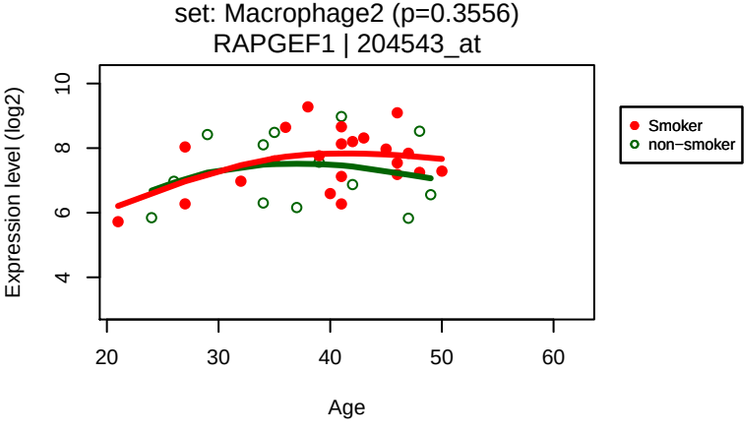

set: SmallAirway2 (p=0.3797)
RAPGEF1 | 204543_at
Expression level (log2)
Age
Smoker
non−smoker

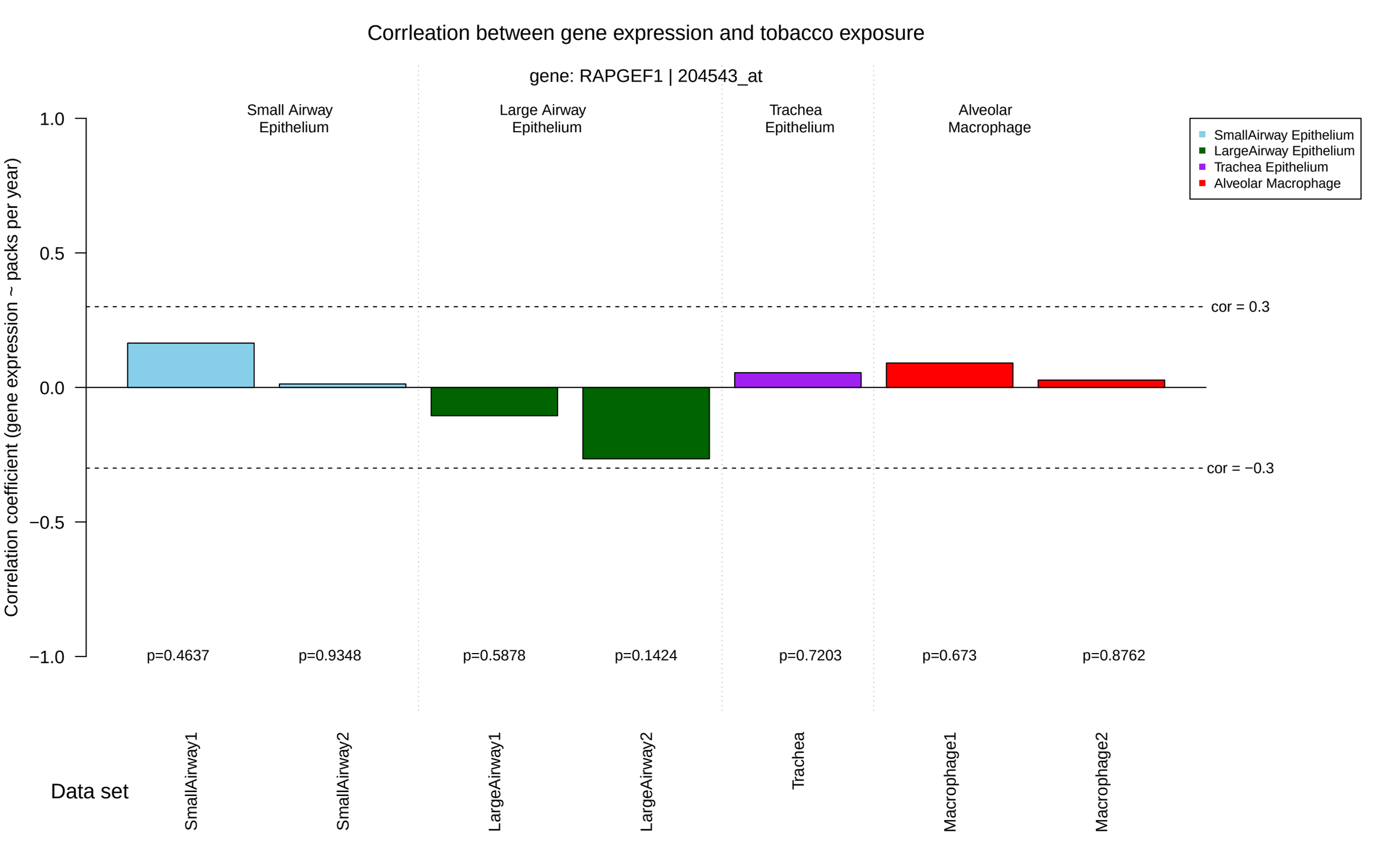

set: LargeAirway1 (p=0.288)
RAPGEF1 | 204543_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9431)
RAPGEF1 | 204543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6873)
RAPGEF1 | 204543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6605)
RAPGEF1 | 204543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3556)
RAPGEF1 | 204543_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAPGEF1 | 204543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4637
p=0.9348
p=0.5878
p=0.1424
p=0.7203
p=0.673
p=0.8762
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2